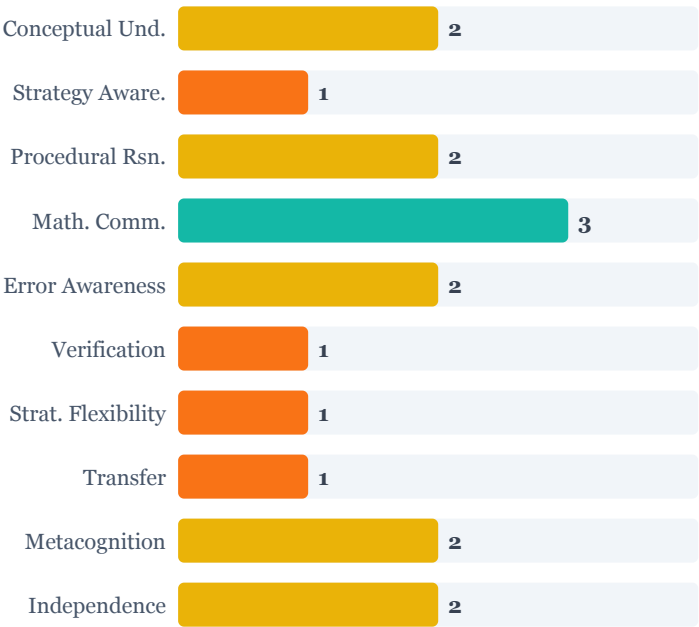


Cayline Yining Cheng

Date of Birth	07 June 2016	School	QSI International Chengdu
Year / Grade	3	Assessment Date	—
Parent / Guardian	Cheng Jin, Liu Chengnan	Referral Reason	Mom will give more information about the reason on the referral form.

RAMRI Disclaimer: RAMRI is a structured qualitative and criterion-referenced reasoning interview. Results must not be represented as standardised scores, age equivalents, grade equivalents, or diagnostic conclusions. This report is a professional qualitative summary and requires interpretation by a qualified educational or psychological practitioner.

Domain Ratings Overview (0 – 4 Scale)



0 No evidence 1 Emerging 2 Developing 3 Demonstrated (limited support) 4 Demonstrated independently

Assessment Context

The RAMRI is a criterion-referenced qualitative interview designed to explore a student's mathematical reasoning through discussion of their own work samples. This session for Cayline Yining Cheng, aged 10 years 1 month, involved discussing seven selected work samples across various mathematical domains to understand her thought processes.

Participation & Emotional Presentation

Cayline engaged with the interview process, demonstrating a degree of confidence and a willingness to articulate her approaches for tasks she perceived as straightforward. For instance, she described identifying vertices as "very easy. You just need to count them" (Sample 1) and acknowledged that "A triangle has three vertices, so it's very simple" (Sample 1). She identified counting circles as potentially challenging because "she might miscount them" (Sample 2). However, for problems she found difficult, such as skip counting by 2s (Sample 3) or missing minuend equations (Sample 6, 7), she often provided no response or explicitly stated "she did not know how to solve the problem." She exhibited good self-correction capabilities when prompted, as seen in Sample 2, where she independently corrected her addition answer. No selection latency or specific selection behaviour was recorded, limiting observations on initial engagement with task selection.

Mathematical Reasoning Profile

Cayline's mathematical reasoning profile, as assessed through the RAMRI, reveals a varied landscape of strengths in procedural execution for familiar tasks and areas for development in conceptual understanding, strategy awareness, and verification.

Her Conceptual Understanding was rated 2/4. This is evidenced by her correct identification and circling of the three vertices of a triangle (Sample 1) and her recognition of addition problems (Sample 2, 5). However, she was "unable to explain what the task required or demonstrate an understanding of counting by twos" (Sample 3) and could not explain the meaning of a minus sign in missing minuend equations (Sample 6, 7).

Her Strategy Awareness was rated 1/4. While she reported consistent strategies like drawing circles for single-digit addition ("this is the method she has always used," Sample 2) and using a number chart for larger additions (Sample 5), she struggled to articulate strategies beyond basic counting for vertices (Sample 1) or shape properties (Sample 4), and stated she "did not know how to solve" missing minuend problems (Sample 6, 7). Procedural Reasoning was rated 2/4. Cayline correctly performed the procedure of circling vertices (Sample 1) and, during the interview, accurately used counting circles for $9+8=17$ (Sample 2) and a number chart for $12+6=18$ (Sample 5). However, her initial written answers contained significant procedural errors, such as $9+8=20$ (Sample 2), incorrect skip counting (2, 4, 6, 8, 10, 11, 12, 13, 14, 15; Sample 3), and $12+6=6$ (Sample 5). Her written work on missing minuend problems (Sample 6, 7) showed attempts at decomposition but no completed answer.

Her Mathematical Communication was rated 3/4, indicating a moderate ability to articulate her thinking for understood tasks, such as explaining her counting method for vertices (Sample 1) and her use of circles or a number chart for addition (Sample 2, 5). Conversely, she provided "No response provided" when asked about conceptual aspects of skip counting (Sample 3) or subtraction (Sample 6, 7). Error Awareness was rated 2/4. Cayline demonstrated capacity for self-correction when prompted, revising $9+8$ from an initial incorrect answer to 17 during the interview (Sample 2). However, she "did not identify any errors" in her written skip counting (Sample 3), her failure to select three shapes (Sample 4), or her incorrect addition (Sample 5). Verification and Reasonableness was rated 1/4. Her primary verification strategy was to "count them again" (Sample 1, 2, 5). She provided "No response provided" for verification strategies for skip counting or missing minuend problems (Sample 3, 6, 7). Strategy Flexibility was rated 1/4, as she used circles for single-digit addition and a number chart for larger addition, but consistently struggled with alternative approaches for other problems.

Her Transfer was rated 1/4, showing limited ability to connect learning from one problem to another, often reiterating the ease of the current task rather than generalisable principles (Sample 1). Metacognition was rated 2/4, evidenced by her comments on task difficulty (Sample 1, 2), but she often provided "No response provided" or no comment on her thinking process for other problems (Sample 3, 4, 5, 6, 7). Finally, Independence was rated 2/4, shown by her self-correction in Sample 2 and her initial attempts at working backwards in Sample 6 and 7, even though she subsequently stated she did not know how to solve them.

Performance vs. Reasoning

Cayline's performance on written tasks sometimes presented a different picture than her verbal explanations. For example, her written answer for $9 + 8$ was "20" (Sample 2), but her verbal explanation and subsequent demonstration using circles yielded the correct answer of "17." Similarly, for $12 + 6$, her written answer was "6" (Sample 5), but during the interview, she correctly used a number chart to find "18." This pattern suggests that while initial written output may reflect procedural errors or lack of focus, her underlying procedural knowledge can be more robust when she verbalizes her process or is guided to demonstrate it. Conversely, for missing minuend problems (Sample 6, 7), her written work showed some partial attempts at decomposition, implying an initial strategy, but she verbally stated she "did not know how to solve the problem," indicating a gap between an attempted procedure and a fully understood and executable solution.

Effect of Assessment Conditions

The qualitative, interview-based nature of the RAMRI, which allows students to discuss their own work, appeared to support Cayline's engagement and reduced some potential for anxiety compared to a typical standardised testing context. Her ability to self-correct the $9 + 8$ problem from "20" to "17" (Sample 2) and to provide clear explanations of her strategies for identified vertices (Sample 1) and addition using a number chart (Sample 5) suggests that the opportunity to verbalize and demonstrate her thinking in a supportive environment enabled her to reveal deeper levels of understanding and problem-solving attempts that might not be evident in solely written, timed assessments.

Domain Coverage

This assessment explored Cayline's reasoning in specific mathematical domains including Geometry, Addition Reasoning, Number Sense, and Subtraction Reasoning. In Geometry, she demonstrated an understanding of vertices and shape properties, although she overlooked instructions in one problem (Sample 1, 4). In Addition Reasoning, she displayed effective counting strategies, particularly when verbally explaining her process (Sample 2, 5). In Number Sense, her conceptual understanding of skip counting by 2s was limited despite some procedural recall (Sample 3). In Subtraction Reasoning, she expressed difficulty with missing minuend problems, despite some written attempts at decomposition (Sample 6, 7). Domains not represented in the submitted work and thus not covered in this report include Multiplicative Reasoning, Division Reasoning, Fractions, Decimals, Percentages, Ratio and Proportional Reasoning, Algebraic Reasoning, Pattern and Relational Reasoning, Mathematical Problem-Solving, Measurement, Spatial Reasoning, Data Interpretation, Statistics, Probability, Money, and Time. The absence of these domains from this report does not indicate difficulty in those areas.

Transferable Reasoning Strategies

Cayline demonstrates several reasoning strengths that are likely to support her learning across unassessed domains. Firstly, her ability to use concrete visual strategies, such as drawing circles for addition (Sample 2) and using a number chart for larger addition (Sample 5), provides a solid foundation. This visual-counting approach can be transferred to other operations or concepts requiring sequential processing, like multiplication (e.g., using arrays) or basic data interpretation (e.g., tallying). Secondly, her capacity for self-correction when prompted, as seen in her independent correction of $9 + 8 = 17$ after miscounting (Sample 2), suggests an underlying ability to monitor her work. This skill is crucial for developing accuracy in complex problem-solving across all domains, from algebraic equations to financial calculations, provided she is taught systematic checking methods. Lastly, her clear Mathematical Communication for concepts she understands (Sample 1, 2, 5) is a valuable strength; being able to articulate her steps allows for easier identification of misconceptions by educators and fosters deeper understanding as she externalizes her thought processes, which is essential for learning new concepts in any mathematical domain.

Strengths Narrative

Cayline demonstrates notable strengths in her ability to communicate her mathematical processes and apply counting-based strategies for addition. For instance, she clearly articulated her method for identifying vertices on a triangle, stating, "This is very easy. You just need to count them" (Sample 1). Her communicative strength is also evident in her explanation of using circles for single-digit addition, stating it's "the method she has always used" (Sample 2), and for larger addition problems, she effectively described using a 1-100 number chart because it allowed her to "count forward more efficiently" (Sample 5). Critically, Cayline exhibited a valuable capacity for error awareness and self-correction. During the interview, when prompted to demonstrate her calculation for $9 + 8$, she realised "she had miscounted the circles" from her initial written answer of "20" and independently corrected it to "17" (Sample 2). This ability to identify and rectify her own errors, particularly in procedural tasks, indicates a strong foundation for developing greater accuracy. Furthermore, her procedural accuracy is evident in her correct identification of vertices in Sample 1 and her accurate execution of addition problems during the interview when applying her counting strategies (Sample 2, 5).

Per-Domain Findings

Geometry

In the domain of Geometry, Cayline demonstrated an understanding of basic geometric properties. For 'Vertices Identification' (Sample 1), she correctly identified and circled all three vertices on a triangle. When asked, she explained, "This is very easy. You just need to count them." She further elaborated that "A triangle has three vertices, so it's very simple," indicating confidence and basic conceptual understanding. For 'Shape Properties (Sides and Vertices)' (Sample 4), Cayline accurately counted the sides of shapes, identifying a square and a trapezoid as having four sides and four vertices. However, she only marked two shapes (square and trapezoid) and did not select the rhombus, which also meets the criteria, thereby failing to follow the instruction to "choose three that apply." She "did not identify any errors in her response and did not recognize that she had overlooked the instruction."

Number Sense

In Number Sense, Cayline demonstrated a procedural recall for the initial part of a sequence but a lack of conceptual understanding. For 'Count by 2s: 2, 4, 6, 8, 10, __, __, __, __' (Sample 3), her written answer was "11, 12, 13, 14, 15." She explained, "I just remembered it: 2, 4, 6, 8, 10, 11, 12, 13, 14, 15." However, when asked about the meaning of counting by 2s, she "was unable to explain what the task required or demonstrate an understanding of counting by twos."

Addition Reasoning

Within Addition Reasoning, Cayline showed a reliance on counting-based strategies. For the problem ' $9 + 8 = \underline{\quad}$ ' (Sample 2), her initial written answer was "20." During the interview, she explained her strategy of drawing circles for each number and then counting them. Upon demonstrating this, she self-corrected her calculation, stating the correct answer was "17" and identifying that "she had miscounted the circles." For ' $12 + 6 = \underline{\quad}$ ' (Sample 5), her written answer was "6." In the interview, she used a 1–100 number chart, located 12, counted forward six numbers, and arrived at "18." She explained that for larger numbers, "drawing circles would require too many representations" and she "preferred using a number chart because it allowed her to count forward more efficiently." In both cases, her verbal reasoning and demonstration revealed more accurate procedural application than her initial written work.

Subtraction Reasoning

For Subtraction Reasoning, particularly involving 'Missing Minuend' problems, Cayline expressed significant difficulty. For ' $___ - 9 = 4$ ' (Sample 6) and ' $___ - 6 = 9$ ' (Sample 7), she stated for both, "she did not know how to solve the problem." She commented, "she could only see the numbers 9 and 4 in the equation" (Sample 6) and "she could only see the numbers 6 and 9 in the equation" (Sample 7). Although her written work for these problems showed partial decomposition (e.g., decomposing 9 into 3 and 6, or 6 into 5 and 1) which indicates an attempt at a strategy, she did not complete the answers or verbally explain her process for these attempts, and provided "No response provided" to questions about conceptual understanding or strategy for these problems.

Strengths

- **Geometry:** Accurate identification of vertices (Sample 1 - "correctly identified and circled all three vertices").
- **Addition Reasoning:** Effective use of visual-counting strategies (Sample 2 - "draws circles to represent each number and then counts"; Sample 5 - "used a 1–100 number chart... counted forward six numbers, arrived at 18").
- **Mathematical Communication:** Clear articulation of known strategies and reasoning for familiar tasks (Sample 1 - "You just need to count them"; Sample 2 - explaining circle drawing; Sample 5 - explaining number chart use).
- **Error Awareness:** Ability to self-correct procedural errors when prompted (Sample 2 - self-corrected $9+8$ from 19/20 to 17 by recounting circles).
- **Procedural Reasoning:** Application of counting strategy to addition tasks yields correct answers during demonstration (Sample 2 - $9+8=17$; Sample 5 - $12+6=18$).

Areas for Development

- **Conceptual Understanding:** Limited understanding of skip counting beyond rote recall (Sample 3 - "unable to explain what the task required or demonstrate an understanding of counting by twos").
- **Conceptual Understanding:** Difficulty with the meaning of operations in less direct contexts (Sample 6, 7 - "No response provided" to "What does the minus sign mean?").
- **Strategy Awareness:** Limited ability to articulate strategies beyond basic counting for tasks like shape properties (Sample 4 - "unable to explain why she chose this approach or describe her strategy beyond counting").
- **Procedural Reasoning:** Inconsistency in initial written procedural accuracy (Sample 2 - $9+8=20$; Sample 3 - shift to counting by 1s after 10; Sample 5 - $12+6=6$).
- **Error Awareness:** Difficulty identifying errors independently without prompting (Sample 3, 4, 5 - "did not identify any errors" or "unable to identify any errors").
- **Verification and Reasonableness:** Limited repertoire of checking strategies, primarily relying on re-counting, and absence of strategies for complex problems (Sample 3, 6, 7 - "No response provided").
- **Strategy Flexibility:** Difficulty applying alternative strategies, particularly for inverse operations or unknown values (Sample 6, 7 - stated "did not know how to solve the problem").
- **Transfer:** Limited ability to connect learning from one problem to general mathematical principles (Sample 1 - response focused on current task ease rather than transferrable learning).
- **Metacognition:** Limited reflection on thinking processes or learning from problems (Sample 4, 5, 6, 7 - "did not comment on the difficulty of the task or describe her thinking process" or "No response provided").
- **Attention to Detail:** Overlooking specific instructions within a problem (Sample 4 - "did not notice that the question required her to 'choose three that apply'").

Strategies for School / Classroom Teachers

- **Conceptual Understanding of Skip Counting:** Provide hands-on activities using manipulatives (e.g., connecting cubes, number line jumps) to physically demonstrate 'groups of two' to build a concrete understanding of counting by 2s, 5s, and 10s (targets Sample 3).
- **Explicit Strategy Instruction and Naming:** Systematically teach and explicitly name different strategies for addition (e.g., 'counting on,' 'making ten,' 'doubles plus one') and subtraction (e.g., 'counting back,' 'counting up,' 'fact families') to expand Cayline's repertoire beyond basic counting (targets Sample 2, 5, 6, 7).
- **Guided Error Analysis:** Regularly incorporate 'spot the error' activities where Cayline and her peers analyze sample work (which may include her own anonymised errors from Sample 2, 3, 4, 5) to identify mistakes and discuss alternative, correct approaches, fostering independent error awareness.
- **Problem Comprehension and Key Word Identification:** Use a visual routine (e.g., 'circle the numbers, underline the question, box the keywords') for word problems and multi-part instructions to ensure Cayline fully processes all elements of a problem statement, such as the 'choose three that apply' instruction in Sample 4.

Strategies for Parents / Home Support

- **Daily Skip Counting Practice:** Engage Cayline in skip counting by 2s, 5s, and 10s in everyday contexts, such as counting socks, coins, or steps, to reinforce the conceptual understanding (targets Sample 3).
- **"Explain Your Thinking" Discussions:** Encourage Cayline to explain how she solved homework problems, asking 'How did you know to do that?' or 'What strategy did you use?' to foster strategy awareness and mathematical communication (targets Sample 1, 2, 5, 6, 7).
- **"Check Your Work" Habit:** When reviewing homework, ask Cayline, 'How could you check if that answer is correct?' or 'Does that answer make sense?' to develop verification strategies beyond simply recounting (targets Sample 1, 2, 5).
- **Careful Reading Practice:** Practice reading instructions together for non-math tasks (e.g., game rules, recipe steps) and discuss the importance of noting all details, then apply this habit to math problems, pointing out specific instructions like 'choose three' (targets Sample 4).

Strategies for Tutors / Learning Support

- **Building Conceptual Bridges for Inverse Operations:** Use concrete manipulatives (e.g., number bonds, counters) and visual models (e.g., part-part-whole mats, number lines) to explicitly teach the relationship between addition and subtraction, focusing on missing minuend problems like Sample 6 and 7 (e.g., for $___ - 9 = 4$, show 4 and 9 coming together to make the whole, rather than just solving $4+9$).
- **Developing Flexible Thinking and Strategy Comparison:** Present problems solvable by multiple strategies (e.g., $12 + 6$ can be 'counting on,' 'making ten with $2+4$,' or 'number chart'). Guide Cayline to explore, articulate, and compare efficiency of different strategies to enhance Strategy Flexibility and Awareness (targets Sample 2, 5).
- **Structured Error Analysis and Verification Routines:** Implement a systematic process for checking answers, moving beyond simple recounting. This could involve using inverse operations (e.g., checking addition with subtraction), estimation (e.g., for $12+6=6$, 'is 6 a reasonable answer if we're adding?'), and self-questioning ('Did I answer all parts of the question?'). This targets Error Awareness and Verification for all samples, especially Sample 2, 3, 4, 5.
- **Metacognitive Questioning for Reflection and Transfer:** Consistently prompt Cayline with questions like 'What was challenging about this problem?' 'What did you learn that you could use next time?' or 'How is this problem like one we've done before?' to foster metacognition and explicit transfer of skills (targets Sample 1, 3, 4, 5, 6, 7).

Recommendations

- Focus on building conceptual understanding of mathematical operations and relationships, particularly for skip counting and inverse operations, through concrete and visual models.
- Encourage Cayline to verbalize her mathematical thinking and strategies for all problems, not just those she finds easy, to help surface misconceptions and build metacognitive awareness.
- Develop a structured approach to problem analysis, emphasizing careful reading of instructions and identifying all parts of a problem before attempting to solve it.
- Explicitly teach and practice a range of verification strategies beyond simple recounting, such as using inverse operations and estimation, to improve error awareness and self-correction.
- Support the development of strategy flexibility by introducing and comparing multiple ways to solve problems, encouraging Cayline to articulate the advantages of different approaches.

Limitations & Caveats

- This is not a standardised assessment.
- The quality of conclusions depends on the authenticity and context of submitted work.
- Previously completed work may have involved unrecorded assistance.
- Transfer evidence is important when interpreting independent understanding.
- Domains not represented in submitted work cannot be assessed — absence from this report does not indicate difficulty in those areas.
- This report requires human review and approval by a qualified professional before release.

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